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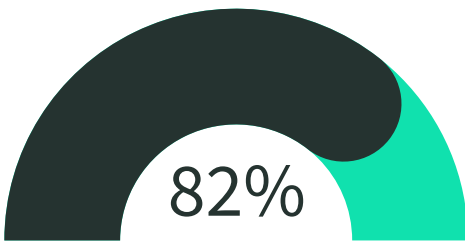
Assi Ghat Boat Electrification Study

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Introduction

Boat rides at Assi Ghat are a vital part of Varanasi's tourism and economy. Traditionally, the fleet of over 800 boats was powered entirely by diesel engines. These boats provide essential jobs for local boatmen.

DATA COLLECTED FROM
PIB GOV OF INDIA



As of 2023, 735 boats out of 890 registered with the Varanasi Nagar Nigam (VNN) had been converted to CNG

Diesel engines have historically caused significant problems, including high fuel costs, expensive maintenance, and environmental damage from noise and pollution. While a recent government initiative has successfully converted over 80% of the fleet to cleaner CNG, challenges related to fossil fuel dependency, engine upkeep, and emissions remain. The move to electric propulsion represents the next logical step towards a truly clean and quiet river experience.

Objective

The main goal of this report is to study if replacing the current internal combustion engines (both diesel and CNG) with electric battery systems is a practical and financially sound decision for the Assi Ghat fleet.

Scope

This is a basic report. It will compare the power needs and running costs for the current boats versus a proposed electric system. It will consider travel with the river's current (downstream) and against it (upstream). This report will not include detailed engineering plans for the conversion.

Current Fleet Analysis

This section details the boats currently operating at Assi Ghat.

- **FLEET COMPOSITION:** The fleet is comprised of three main boat types:
 - (i). 10 HP Standard Boat: Carries 5-10 passengers. This is the most common type.
 - (ii). 20 HP Mid-Size Boat: Carries up to 20 passengers.
 - (iii). 180 HP Large Vessel: A cruiser for large groups of 100+ people.

OPERATION COST FOR A 10 HP BOAT

A typical tourist round trip from Assi Ghat to Manikarnika Ghat and back (approx. 7 km total) takes about 1.5 hours.

- **Fuel:** A typical 1.5-hour round trip to Manikarnika Ghat costs approximately ₹204 with **CNG** [1.5 hours * 1.6 kg/hr * ₹85/kg] and with **diesel** it is ₹270 [1.5 hours * 2.0 L/hr * ₹90/L]
- **Maintenance:** Engine service is required every 80 hours, leading to an estimated annual cost of ₹27,000.
Estimated cost per service is ₹2000
Assuming two trips a day of 3 hours this is roughly 27 days
Hence Annual cost = $365/27 * ₹2000 = ₹27,000$

Proposed Electric Propulsion System

System Overview

The proposed system replaces the combustion engine with four main components:

1. **Electric Motor:** A quiet, efficient motor to drive the propeller.
2. **Battery Bank:** A large battery pack to store energy, typically using Lithium-ion (Li-ion) or Lithium Ferro Phosphate (LFP) technology.
3. **Motor Controller:** The "brain" that manages power from the batteries to the motor.
4. **Charging System:** An onboard or shore-based system to recharge the batteries.

Assumptions for Analysis

The following values are used for our calculations:

1. Average Boat Speed: 8 km/h.
2. Ganga Current Speed: 2 km/h (average).
3. Daily Operating Time: 4 hours (two round trips).
4. Electricity Cost: ₹10 per kWh (kilowatt-hour) .
5. Water Density (ρ): 1000 kg/m³ .

Technical Feasibility Analysis

Power Requirement

Power needed to propel the boat is directly proportional to the cube of boats speed relative to water
 $p \propto v^3$

Upstream effective speed: 8 km/h
 (boat) + 2 km/h (current) = **10 km/h**

Downstream effective speed: 8 km/h
 (boat) - 2 km/h (current) = **6 km/h**

Boat Type	Existing HP	Equivalent kW	Recommended Electric Motor
Standard	10 HP	7.5 KW	10 KW
Mid-Size	20 HP	15 KW	20 KW
Large	180 HP	134 KW	150 KW

Battery Sizing Calculation

- The battery must store enough energy for a full day (4 hours)
- Downstream Power Formula: $P_{down} = P_{up} \times (v_{down}/v_{up})^3 = P_{up} \times (6/10)^3 = P_{up} \times 0.216$
- we calculate the total daily energy usage for 2 hours upstream and 2 hours downstream.
- Energy Formula: $E_{daily} = (P_{up} \times 2 \text{ hrs}) + (P_{down} \times 2 \text{ hrs})$
- Battery Size Formula: $\text{Size} = E_{daily} / 0.8$ (assuming 80% efficiency)

Boat type	Upstream Power (P _{up})	Downstream Power (P _{down})	Daily Energy (E _{daily})	Required Battery Bank
standard	7.5 KW	1.62 kW (7.5 * 0.216)	18.24 kWh (15 + 3.24)	30 kWh (22.8 kWh needed, 30 kWh is a safe, standard size)
mid sized	15 KW	3.24 kW (15 * 0.216)	36.48 kWh (30 + 6.48)	60 kWh (45.6 kWh needed, 60 kWh is a safe, standard size)
large	134 KW	28.9 kW (134 * 0.216)	325.8 kWh (268 + 57.8)	400 kWh (407 kWh needed, 400 kWh is a suitable target size)

Financial Analysis & Comparison

This includes the motor, batteries, controller, and installation. Marine-grade electric systems are costly.

INITIAL INVESTMENT

Boat Type	Motor Size	Battery Size	Estimated Initial Investment
Standard	10 kW	30 kW	₹7,00,000
Mid-Size	20 kW	60 kWh	₹13,00,000
Large	150 kW	400 kWh	₹80,00,000

OPERATIONAL COST PER DAY (FOR STANDARD BOAT)

Cost Item	CNG System	Electric System	Daily Savings
Fuel	₹408 (2 trips)	₹168 (16.8 kWh * ₹10)	₹240
Maintenance	₹74 (₹27k / 365)	~₹10 (minimal)	₹64
Total	~₹482	~₹178	~₹304

PAYBACK PERIOD

Annual Savings: ₹304/day * 365 days = ₹1,10,960

Payback Period: ₹7,00,000 (Investment) / ₹1,10,960 (Annual Savings)
 ≈ **6.3 years**

Challenges and Practical Considerations

- 1. High Initial Cost:** The upfront investment is the single largest barrier for boatmen. Government subsidies or financing schemes would be essential.
- 2. Charging Infrastructure:** A robust network of charging stations would need to be built at the ghats. This is a significant infrastructure project.
- 3. Battery Availability & Lifespan:** Sourcing reliable, marine-grade LFP batteries is crucial. These batteries have a lifespan of 8-10 years, and their eventual replacement cost must be factored into the long-term financial plan.
- 4. Charging Time:** A 30 kWh battery could take 4-5 hours to charge with a standard fast charger, meaning boats would have significant downtime.
- 5. Training:** Boatmen would require training on the operation and basic maintenance of the new electric systems.

Conclusion and Recommendations

Summary

The conversion of the Assi Ghat fleet to electric propulsion is technically feasible. The long-term operational savings on fuel and maintenance are significant. However, the high initial investment presents a major financial hurdle, with a payback period exceeding 6 years for a standard boat.

Recommendations

- 1. Pilot Program:** It is strongly recommended to launch a government-supported pilot program by converting 5-10 boats. This would provide real-world data on performance, costs, and charging needs.
- 2. Seek Subsidies:** The government should explore subsidies under national clean energy or tourism initiatives to help boatmen overcome the initial investment barrier.
- 3. Infrastructure Planning:** A detailed study should be commissioned to plan the development of a scalable charging infrastructure at Assi Ghat and other key locations.